

Day 3: Random Variables and Distributions

Mathcamp 2026 — Estimation Theory

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From Measurements to Randomness

So far, we've treated data points as fixed numbers. But real measurements have *randomness* — repeat an experiment and you'll get different values. Today we build the mathematical framework to describe this randomness.

Discrete Random Variables

A Loaded Die

Suppose someone hands you a six-sided die that might be loaded. You roll it 600 times and observe:

Outcome	1	2	3	4	5	6
Count	85	92	103	108	112	100

Exercise 1: Discrete Probability

- (a) Based on this data, estimate the probability of rolling each outcome.
- (b) A fair die has $P(X = k) = 1/6$ for each outcome. Does this die appear fair?
- (c) What is your estimate for $P(X \leq 3)$? For $P(X \text{ is even})$?
- (d) If you rolled this die 1200 times instead, would your probability estimates change? Would your *confidence* in them change?

💡 Definition: Discrete Random Variable

A **discrete random variable** X takes values in a finite or countable set. Its **probability mass function (PMF)** is $p(x) = P(X = x)$, satisfying:

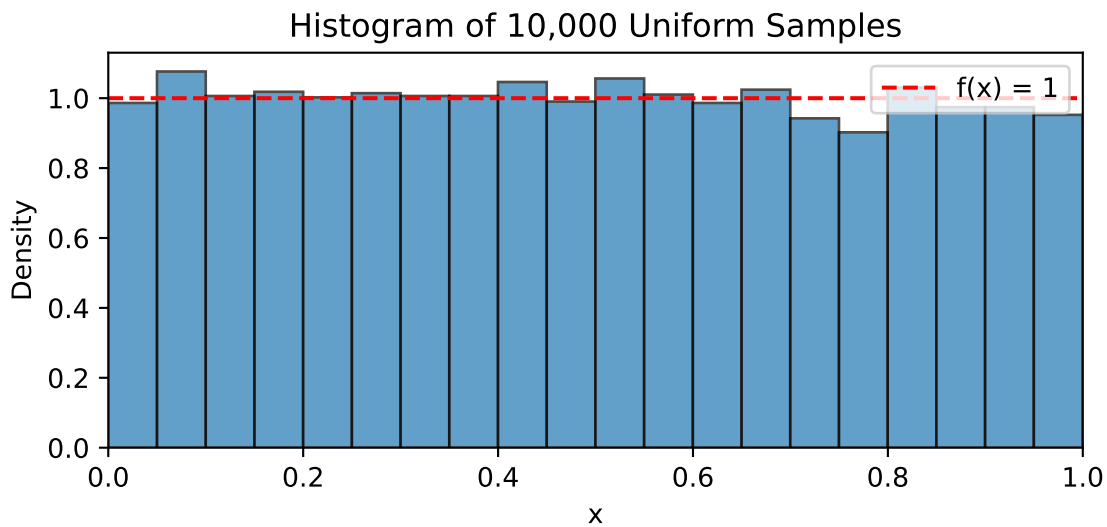
- $p(x) \geq 0$ for all x
- $\sum_x p(x) = 1$

Continuous Random Variables

Many quantities — time, distance, temperature — take continuous values. We can't assign probability to each individual value (there are uncountably many!), but we can assign probability to *intervals*.

i Exercise 2: From Histograms to Density

A random number generator claims to produce values uniformly in $[0, 1]$. You generate 10,000 samples and create a histogram:



i Note

- (a) The histogram is normalized so that area = 1. Why is this the right normalization?
- (b) The red dashed line shows $f(x) = 1$. What does the height represent? (Hint: it's not probability!)

(c) Estimate $P(0.3 \leq X \leq 0.5)$ from the histogram. How does this relate to area?

💡 Definition: Continuous Random Variable

A **continuous random variable** X has a **probability density function (PDF)** $f(x)$ satisfying:

- $f(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f(x) dx = 1$

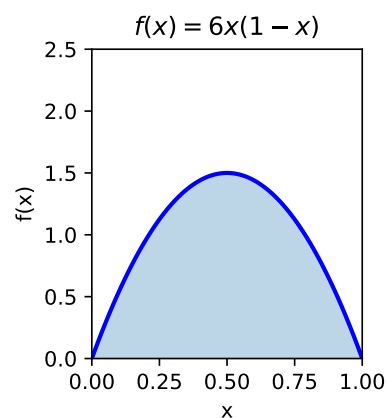
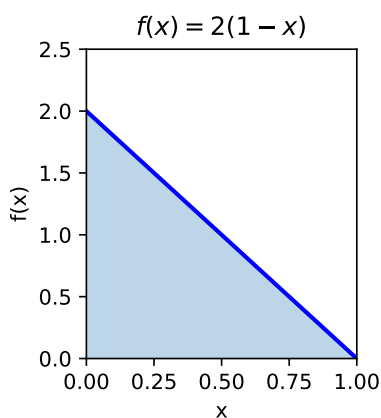
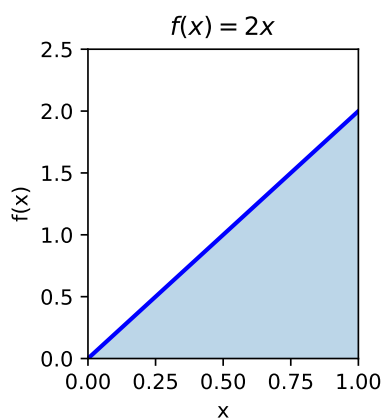
Probabilities are areas: $P(a \leq X \leq b) = \int_a^b f(x) dx$

Non-Uniform Distributions

i Exercise 3: Working with Densities

Consider these density functions on $[0, 1]$ (and zero elsewhere):

- $f(x) = 2x$: Verify this is a valid PDF. Find $P(X > 0.5)$ and $P(0.25 \leq X \leq 0.75)$.
- $f(x) = 2(1 - x)$: Verify this is a valid PDF. Find $P(X < 0.5)$.
- Compare your answers to (a) and (b). What's the relationship between these distributions?
- $f(x) = 6x(1 - x)$: Verify this is a valid PDF. Find $P(0.25 \leq X \leq 0.75)$.



💡 The Beta Family

These densities are special cases of the **Beta distribution**: $f(x) \propto x^a(1-x)^b$ for $a, b > -1$. The cases above correspond to small integer values of a and b . This family will reappear when we study estimation.

Independence

When are two random events unrelated?

i Exercise 4: Independent or Not?

For each scenario, let A and B be the indicated events. Decide whether knowing A occurred changes the probability of B .

- (a) Flip two fair coins. A = “first coin is heads”, B = “second coin is heads”
- (b) Roll two fair dice. A = “first die shows 6”, B = “sum is greater than 7”
- (c) Draw two cards from a standard deck *without* replacement. A = “first card is an Ace”, B = “second card is an Ace”
- (d) Draw two cards *with* replacement. A = “first card is an Ace”, B = “second card is an Ace”

💡 Definition: Independence

Events A and B are **independent** if $P(A \cap B) = P(A) \cdot P(B)$.

Random variables X and Y are **independent** if for all values x, y :

$$P(X = x \text{ and } Y = y) = P(X = x) \cdot P(Y = y)$$

For continuous variables, this becomes: $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$

i Exercise 5: Testing Independence

- (a) You flip a fair coin twice. Let X = number of heads on first flip, Y = number of heads on second flip. Verify that X and Y are independent by checking the factorization for all four combinations of values.
- (b) You flip a fair coin twice. Let X = number of heads on first flip, Z = total number of heads. Are X and Z independent? Check the factorization.

- (c) (X, Y) is uniformly distributed on the unit square $[0, 1]^2$. This means $f_{X,Y}(x, y) = 1$ for $0 \leq x, y \leq 1$. Are X and Y independent?

Multiple Independent Measurements

Here's why independence matters for estimation: when we make *repeated measurements* of the same quantity, we typically assume the measurement errors are independent.

i Exercise 6: Combining Measurements

You measure a length three times with independent uniform errors. Each measurement X_i is uniformly distributed on $[\theta - 0.5, \theta + 0.5]$ where θ is the true length.

- What is the PDF of each X_i ?
- Write down the joint PDF $f(x_1, x_2, x_3)$ using independence.
- If you observe measurements $x_1 = 5.2$, $x_2 = 4.9$, $x_3 = 5.1$, what values of θ are *consistent* with all three measurements? (That is, for what θ is $f(x_1, x_2, x_3) > 0$?)
- This is a preview of tomorrow's topic. We'll learn how to find the *best* estimate of θ given the data.

Summary

Today we established the probabilistic framework for estimation:

Concept	Discrete	Continuous
Probability function	PMF: $p(x) = P(X = x)$	PDF: $f(x)$
Normalization	$\sum_x p(x) = 1$	$\int f(x) dx = 1$
Computing probabilities	Sum the PMF	Integrate the PDF
Independence	$P(X = x, Y = y) = P(X = x)P(Y = y)$	$f_{X,Y}(x, y) = f_X(x)f_Y(y)$

Key insight: Independence lets us write joint distributions as products, which simplifies the mathematics of combining multiple measurements.

Looking Ahead

Tomorrow: Given observed data $x_1, \dots, x_n$ from a distribution that depends on an unknown parameter θ , how do we find the “best” estimate of θ ? The answer is **Maximum Likelihood Estimation**, which combines everything we’ve learned.